

Skill Enhancement Courses (SEC) for First, Second, Third & Fourth Semester

S.No	Course Code	Course Title	Credit Hours	Remarks
1	AEN 121	Bee Keeping	2 (0+2)	- Students can select any one course based on their interest or the course will be allotted based on the +2 Aggregate marks - Minimum number of students required to offer a course will be 20 and maximum number per course will be 40 students - Maximum of four SEC courses may be taken from any department, subject to a limit of one SEC course per department in each semester.
2	AEN 122	Commercial Sericulture	2 (0+2)	
3	AGR 121	Organic inputs preparation techniques	2 (0+2)	
4	SAC 121	Soil, Plant and Water Testing	2 (0+2)	
5	FSN 121	Bakery Technology	2 (0+2)	
6	FSN 122	Processing and Value-addition of Fruits and Vegetable Products	2 (0+2)	
7	HOR 121	Nursery Production in Horticulture Crops	2 (0+2)	
8	HOR 122	Orchard Floor Management	2 (0+2)	
9	HOR 123	Landscape Gardening	2 (0+2)	
10	ARM 121	Entrepreneurial Skills for Agribusiness	2 (0+2)	
11	GPB 121	General morphology and Pollination techniques in Agricultural crops	2 (0+2)	
12	SST 121	Seed production technology	2 (0+2)	
13	AMP 121	Broiler Production	2 (0+2)	
14	PAT 121	Commercial Mushroom Production	2 (0+2)	
15	COM 124	Database Management and its Application in Agriculture	2 (0+2)	
16	*CRP 121	Diagnosis and Mitigation of Physiological Disorders in Crops	2 (0+2)	
17	*AGM 121	Microbial products	2 (0+2)	
18	*ENS 121	Value added production technologies for waste utilization	2 (0+2)	

*** The syllabus will be circulated after obtaining the approval of the 77th Deans' Committee.**

AEN 121 Bee Keeping (0+2)

Practicals

Unit I: Bee species, castes and social life

Honeybee species, castes and biology; morphology and structural adaptations; behaviour; communication; colony organization; pollen and nectar pasturage crops; role of honey bees in crop pollination

Unit II: Bee Keeping equipment's and apiary establishment

Bee hives and bee keeping appliances; comb foundation mill; visit to bee keeping equipment manufacturing unit; site selection and establishment of a Apiary; hive inspection and maintenance of hive records; Meliponiculture; hiving Indian bee and stingless bee feral colony

Unit III: Apiary maintenance

Dividing and uniting of honey bee colonies; artificial feeding of nectar and pollen; seasonal management; management of swarming, absconding and laying workers; pests and diseases of honey bees and their management; protecting bees from pesticides, migratory bee keeping,

Unit IV: Bee products

Honey extraction, processing, purity testing and value addition, mass queen rearing, collection of pollen, bee venom and other bee products and their uses.

Unit V: Bankable project

Visit to Honey FPOs; Preparation of bee keeping projects for bank funding and its economics

Practical Schedule

1. Identification of Honey Bee Species, Bee castes and Biology
2. Honey bee morphology and structural adaptations
3. Bee Behaviour
4. Bee communication – dance and pheromones
5. Colony Organization in Honey Bees
6. Pollen and nectar pasturage crops in bee keeping
7. Preparation of floral calendar
8. Role of Honey Bees in Crop Pollination
9. Bee Hives and bee keeping appliances
10. Production Protocol of Comb Foundation Sheet in Comb Foundation Mill
11. Visit to bee keeping equipment manufacturing unit
12. Site selection and establishment of a Apiary
13. Hive inspection and maintenance of hive records
14. Hiving Indian bee feral colony

15. Meliponiculture
16. Hiving Stingless bee feral colony
17. Artificial feeding of nectar and pollen during dearth period
18. **Mid Semester Examination**
19. Dividing and Uniting of honey bee colonies
20. Seasonal Management of Honey Bees
21. Management of swarming, absconding and laying workers
22. Insect pests of honeybees and their management
23. Non-Insect pests of honeybees and their management
24. Identification of diseases of honeybees and their management
25. Bee poisoning and management
26. Migratory bee keeping and visit to migratory bee keeping site
27. Honey extraction and processing
28. Purity testing and value addition of honey
29. Bee pollen, bee venom (Apitherapy) – collection and uses
30. Other bee products- Bee wax, propolis and royal jelly – collection and uses
31. Mass queen rearing techniques
32. Visit to Honey FPOs
33. Preparation of bee keeping projects for bank funding and its economics
34. **Final Practical Examination**

References

1. Abrol D.P. 2011. Beekeeping – A comprehensive guide to bees and beekeeping. Scientific Publishers, Jodhpur, 896 pages
2. Atwal, A.S.2013.Mellifera Bee Keeping & Pollination. KalyaniPublishers, Ludhiana. 394 p.
3. Ted Hooper, 1991. Guide to Bees and Honey (Thrid Edition), BASprintersLtd. OverWallop, Hampshire271 p.
4. Roger A. Morse, 1994. The new complete guide to beekeeping. *The CountrymanPress*, Woodstock, Vermont. 207p.
5. Thomas D.Seeley.1995.The Wisdom of the Hive, Harvard University Press, Cambridge, 295p.
6. $\square\square.\square\square.\square\square\square\square\square\square\square$, $\square.\square\square\square\square\square$, $\square.\square\square\square\square\square$ $\square\square\square\square\square\square\square\square\square$. 2025. $\square\square\square\square$ $\square\square\square\square$ $\square\square\square\square$ $\square\square\square\square\square\square\square\square$ $\square\square\square\square\square\square\square\square\square$ $\square\square\square\square\square\square\square\square$ $\square\square\square\square$ $\square\square\square\square$ $\square\square\square\square\square\square\square\square$ $\square\square\square\square\square\square\square$: $\square\square\square$ $\square\square\square\square\square\square\square$ $\square\square\square\square\square\square\square$, $\square\square\square\square\square$ 246 $\square$. ($\square\square\square\square$)

e- Resources

1. http://agritech.tnau.ac.in/farm_enterprises/fe_apiculture_home.html
2. <http://agdev.anr.udel.edu/maarec/>
3. <http://www.aragriculture.org/insects/beekeeping.htm>
4. <https://scientificbeekeeping.com>
5. <http://beekeeping.com/>; <http://www.apimondia.com/en>

AEN 122 Commercial Sericulture (0+2)

Practicals

Unit I : Mulberry and its cultivation

Mulberry varieties - Mulberry nursery techniques, Preparation of land, planting material, planting of mulberry, Pruning, harvesting and storage of mulberry – Mulberry pests and diseases

Unit II : Biology of mulberry silkworm

Identification of cocoons of important breeds and cocoons - External morphology of life stages egg, larva, pupa and moth of *Bombyx mori* L - Study of silk glands and digestive system of *Bombyx mori* -Pests and diseases of mulberry silkworm

Unit III : Silkworm rearing and infrastructure

Disinfectants - general and rearing bed disinfectants – silkworm seed production techniques - Study of rearing house plan, structure and cost estimates - equipments for shoot feeding and shelf rearing - Incubation and black boxing of silkworm eggs – different methods - brushing of larvae- Identification of moulting of silkworm settling – Feeding - Bed cleaning and bed spacing - Identification of ripened silkworm – Mounting process - Types of mountages

Unit IV : Cocoon harvest and reeling

Cocoon harvesting, sorting, grading and marketing– Study of reeling equipments – Charkha, Multiend reeling machine and Automatic Reeling unit – Economics of mulberry silk production

Unit V : Non mulberry silkworm & Practical exposure

Different host plants of non-mulberry silkworms - Biology and rearing techniques of Eri, Tasar and Muga silkworms - By-products utilization and value addition - Visits to CRC, Mulberry garden and silkworm rearing units, silkworm seed production centre and Reeling units.

Practical Schedule

1. Mulberry –important varieties, salient characters and yield – irrigated and rainfed varieties
2. Nursery technology - selection of planting material, Azospirillum and VAM treatment, planting, crop protection and uprooting
3. Mulberry garden – Establishment and maintenance – field preparation, planting methods, pruning methods, fertilization, irrigation, leaf and shoot harvest
4. Identification of insect pests of mulberry, damage symptoms and management
5. Identification of diseases and nematodes of mulberry, causal organism, damage symptoms and management

6. Classification and identification of different silkworm breeds and cocoons
7. External morphology of different life stages of *Bombyx mori* - egg, larva, pupa and moth
8. Study of silk glands of *B.mori*
9. Study of digestive system of mulberry silkworm
10. Identification of pests of mulberry silkworm, symptoms and management
11. Identification of diseases of mulberry silkworm, symptoms and management
12. Disinfection - General and Rearing bed disinfectants –Preparation of disinfection solution and disinfection of rearing house and appliances
13. Silkworm rearing house – Plan, structure, types, Environmentally Controlled (EC) rearing house and cost estimates
14. Silkworm rearing appliances for shoot feeding and shelf rearing
15. Incubation & black boxing of silkworm eggs-different methods; brushing of silkworm larvae
16. Chawki rearing techniques and Chawki Rearing Centre
17. Late age silkworm rearing – Floor, Tray and Shoot rearing methods

18.Mid Semester Examination

19. Identification of moulting and moulting care – bed disinfection
20. Feeding of silkworm larvae, bed cleaning and bed space maintenance
21. Identification of ripened larvae, mounting techniques and types of mountages
22. Cocoon harvesting, sorting, grading, transportation and marketing
23. Silkworm seed production techniques – Production of card/sheet egg and loose eggs
24. Study of reeling equipments-Charkha, Multiend reeling machine and Automatic Reeling unit
25. By-products utilization and value addition in Sericulture
26. Economics of mulberry silk production
27. Study of host plants of non-mulberry silkworm –Eri, Tasar and Muga
28. Biology of Eri, Tasar and Muga silkworms- Egg, Larva, Pupa and Adult
29. Rearing techniques of Eri, Tasar and Muga silkworms
30. Visit to Chawki Rearing Centre (CRC) to study the chawki rearing techniques and record maintenance
31. Visit to progressive farmer's field - Mulberry garden and silkworm rearing unit

32. Visit to Silkworm Seed Production Centre

33. Visit to silk reeling unit

34. **Final Practical Examination**

References

1. Dandin, S.B. and Giridhar, 2010, Handbook of Sericulture Technologies. CSB, Bengaluru, P.427.
2. Ganga, G. and SulochanaChetty, J., 1991, An Introduction to Sericulture. Oxford and IBH, New Delhi, P. 176.
3. Govindan, R., Narayanaswamy, T.K. and Devaiah, M.C., 1998, Principles of Silkworm
4. Pathology. Seri Scientific Publishers, P. 420.
5. Rangaswami, G., Narasimhanna, M.N., Kasivishwanathan, K., Sastry, C.S. and Jolly, M.S., 1979, Mulberry cultivation. FAO Manual, Vol. I, FAO, Rome, pp. 99-129.
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9. Nataraju, B., Sathyaprasad, K., Manjunath, D. and Aswani Kumar, C., 2005, Silkworm Crop Protection. Central Silk Board, Bengaluru, pp.1 – 324.
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e- resources

1. www.csb.gov.in/
2. www.tnau.ac.in/
3. www.csrtimys.res.in/

AGR 121 Organic Inputs Preparation Techniques (0+2)

Practicals

Unit I: Organic Nutrient Sources and Composting

Quantification of cow dung and cow urine recovery per animal; nutrient composition analysis; Assessment of storage methods; Collection, segregation, shredding, and quantification of farm wastes for compost preparation; pre-digestion and preparing bio-compost and vermicompost; FCO standards; Enriched organic manure/compost preparation; Compost maturity of FAO standards.

Unit II: Green Manures and Organic Formulations

Quantification of biomass for different green manures and green leaf manures; multivarietal seed technique; Preparing liquid formulations of Panchagavya, fermented extracts from various sources and Effective Microorganisms (EM).

Unit III: Biofertilizers and Bioinoculants

Mass multiplication and use of PPFM and liquid bioinoculants; Mass multiplication and use of bioinoculants for crop application; Mass production of Arbuscular Mycorrhizal Fungi (AMF); Biodiversity assessment techniques

Unit IV: Weed and Pest Management in Organic Farming

Integrated organic weed and pest management; Experiencing mulching techniques in weed management; Preparation and application of herbal leaf and fermented extracts in pest and disease management; Monitoring of insect pests through traps and lures; Biodiversity assessment techniques

Unit V: Organic Certification and Post-Harvest Management

Organic certification procedures – importance, scope, and documentation; Post-harvest management, value addition, and supply chain management in organics; Role of FPOs.

Practical Schedule

1. Quantification of cow dung and cow urine recovery per animal
2. Quality and nutrient composition analysis of cow urine and cow dung
3. Assessment of storage methods with temperature, nutrient, and microbial load in manure pits.
4. Collection, segregation, shredding and quantification of farm wastes
5. Pre-digestion of various farm wastes for composts preparation
6. Preparing bio-compost from farm wastes and quality increase to match FCO standards

7. Preparing vermicompost
8. Assessing the quality of various composts
9. Enriched organic manure/compost preparation
10. Harvest and assessment of compost maturity indices against FAO standards.
11. Quantification of biomass for different green manures and green leaf manures
12. Nutrients composition analysis of green manures and green leaf manures
13. Multi-varietal seed technique (Navadhanya) for soil quality improvement
14. Preparing the liquid formulations of *Panchagavya*, and effective microorganisms
15. Preparing the fish and egg fermented extracts
16. Mass multiplication and use of biofertilizers, Liquid bioinoculants and PPFM
17. Mass production of Arbuscular Mycorrhizal Fungi
18. **Mid Semester Examination**
19. Insitu (Field) soil health assessment techniques
20. Exposure visit to organic farming farmer's field
21. Integrated organic weed management techniques
22. Experiencing mulching techniques in weed management
23. Integrated organic pest management
24. Preparation and application of herbal leaf extracts in insect management
25. Preparation and application of herbal /fermented extracts in disease management
26. Monitoring of insect pests through traps and lures
27. Importance and establishment of pest repellent cafeteria
28. Biodiversity assessment techniques
29. Organic certification procedures
30. Post-harvest management and value addition of organic produces
31. Supply chain management in organic agriculture
32. Role of FPOs in organic agriculture, Opportunities for entrepreneurship development
33. Exposure visit to TNOCD unit/FPOs / Organic inputs producing industries
34. **Final Practical Examination**

References

1. M. Suganthi B. Renuka Rani N R Sharma. 2023. Natural Farming for Sustainable Agriculture. E-book published by Hyderabad: National Institute of Agricultural Extension Management (MANAGE). ISBN: 978-81-19663-10-1
2. S.S. Walia, Tamanpreet Kaur & N Ravisankar. 2025. Principles and Practices of Natural Farming. Scientific Publishers. ISBN9789363461079.

3. Palaniappan, S. P. (2018). *Organic farming: Theory and practice*. Scientific Publishers. ISBN: 978-8172335380. <https://www.scientificpubonline.com/bookdetail/organic-farming-theory-practice/9788172335380/0>
4. Suganthi, M., P. Janaki, R. Krishnan, P. Balasubramaniam, S. Lakshmi, S. Susikaran, P. Kumaresan. 2025. *Organic Farming*. Directorate of Open and Distance Learning, TNAU, Coimbatore. TNAU Offset Press, 56p, ISBN. 978-93-93563-41-5.
5. M. Suganthi, P. Janaki, E. Parameswari, R. Krishnan, M. Ramasubramanian, P. S. Kavitha, and M. K. Kalarani. 2023. "Natural Farming: present status and Future prospectus. Published by Darshan Printers, 6A, Pudhukinar Street, P.N. Pudur, Coimbatore. ISBN. 978-93-92032-93-6. P. 487.
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11. Ramesh, C. (2021). *Handbook of organic farming and certification*. New India Publishing Agency. ISBN: 978-9388892124.
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13. Sharma, S. K. (2018). *A handbook of organic farming*. Agro-Bios. ISBN: 978-8177540994.
14. Kristiansen, P. (2020). *Organic farming: Science and practice*. Acram Publishers. ISBN: 978-9388892124.
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1. <https://www.fao.org/organicag/oa-faq/oa-certification/en/>

2. <https://www.amazon.com/Organic-Farming-Science-Practice-Kristiansen/dp/938889212X>
3. <https://agrobiosdigitals.com/ebookchapters.php?bx===AUVZ0cWtGZXJFbaNVTWJVU&ChapterView=1>
4. https://www.researchgate.net/publication/387819808_Integrated_Insect_Pest_Management_in_Organic_Farming
5. <https://www.biblio.com/book/organic-farming-farming-systems-sustainable-agriculture/d/1614935503?srsItd=AfmBOorlC4QeHU6d7Iab0D424eHSgum9gkpz8copVd3Lh5LaevbCiQ0Y>
6. <https://www.abebbooks.com/9789381095140/Composting-Organic-Fertilizers-Malik-R-9381095146/plp>
7. <https://www.satishserial.com/book/9789388892117/organic-farming-and-farming-systems-for-sustainable-agriculture>
8. <https://archive.org/details/biofertilizersin0000subb>
9. <https://www.walmart.com/ip/Soil-Fertility-and-Nutrient-Management-Principles-and-Practices-Hardcover-9781032429298/1118374774>

SAC 121 Soil, Plant and Water Testing (0+2)

Practicals

Unit I: Collection of soil sample

Collection of representative soil sample- processing and storage. Determination of soil density - bulk density (wax coating method) and particle density (cylinder method), Determination of Soil moisture content. Determination of soil colour. Determination of soil texture by feel method and Bouyoucos Hydrometer method. Determination of Hydraulic conductivity.

Unit II: Preparation and standardization of laboratory reagents

Preparation and standardization of laboratory reagents: acidimetry, alkalimetry. Preparation of indicators and buffers. Determination of soil pH and electrical conductivity. Determination of cation exchange capacity of soil. Estimation of soil organic Carbon. Determination of lime requirement of acid soil. Determination of gypsum requirement of sodic soil. Estimation of soil available Macronutrients (N, P and K), secondary nutrients (Ca, Mg and S) and Micronutrients.

Unit III: Plant analysis

Plant analysis: collection of plant samples and processing. Digestion of plant sample - preparation of diacid and triacid mixture. Estimation of total nutrients in plant samples: nitrogen, phosphorus, potassium, calcium and magnesium, sulphur and micronutrients.

Unit IV: Irrigation water

Irrigation water: collection of different types of irrigation water samples. Irrigation water analysis: estimation of pH and EC, TSS and hardness of irrigation water. Estimation of cations: calcium, magnesium, sodium and potassium. Estimation of anions in irrigation water: carbonate and bicarbonates, chlorides sulphate and boron. Estimation of Micronutrients (Mn, Fe, Zn and Cu). Classification of irrigation waters as per USSS and other systems. Computation of salts (RSC and SAR), Water quality indices and appraisal.

Unit V: Interpretation

Interpretation of soil, plant and water test results with crop response; Soil Test based recommendations, DISSIFER, VDK, Preparation of soil health cards. Scope of establishing soil testing and water testing laboratories. Accreditation and quality control measures. Entrepreneurship opportunities in soil, plant and water testing advisory services in agriculture. Visit to soil testing laboratory (STL).

Practical Schedule

1. Collection of representative soil, plant and water samples- processing and storage
2. Determination of soil density and interpretation

3. Determination of Soil moisture, colour and their interpretation & soil texture by feel method and Bouyoucos Hydrometer method
4. Determination of Hydraulic conductivity and ratings
5. Preparation and standardization of laboratory reagents: acidimetry, alkalimetry
6. Preparation of indicators and buffers. Determination of soil pH and electrical conductivity and interpretation
7. Determination of cation exchange capacity of soil and interpretation
8. Estimation of organic carbon content of soil and ratings
9. Determination of lime requirement of acid soil & gypsum requirement of sodic soil - interpretation and reclamation
10. Estimation of soil available Nitrogen
11. Estimation of soil available Phosphorus (Bray and Olsen' P) and Potassium
12. Estimation of Exchangeable Ca and Mg and soil available S
13. Estimation of DTPA Extractable Micronutrients
14. Digestion of plant sample - preparation of diacid and triacid mixture: Estimation of total nutrients in plant samples: Nitrogen
15. Estimation of total nutrients in plant samples :Phosphorus and Potassium
16. Estimation of total nutrients in plant samples: Calcium and magnesium
17. Estimation of total nutrients in plant samples: Sulphur and micronutrients.
18. **Mid Semester Examination**
19. Rapid tissue tests for plant nutrient analysis
20. Irrigation water analysis- estimation of pH and EC, TSS, hardness and rating
21. Irrigation water analysis: estimation of cations (Calcium and Magnesium) and rating
22. Irrigation water analysis: estimation of cations (Sodium and Potassium) and rating
23. Irrigation water analysis: estimation of anions (Carbonate and bicarbonates) and rating
24. Irrigation water analysis: estimation of anions (Chlorides) and rating
25. Irrigation water analysis: estimation of anions (Sulphate) and rating
26. Irrigation water analysis: estimation of Micronutrients (Mn, Fe, Zn, Cu & B) and rating
27. Classification of irrigation waters as per USSL and other systems.

28. Computation of salts, Water quality indices and appraisal
29. Analytical advisory services : Soil test based recommendations, DISSIFER & VDK
30. Preparation of soil health cards and recommendations
31. Establishing Startups - soil and water testing - Accreditation and quality control measures
32. Entrepreneurship opportunities in soil testing advisory services-
33. Visit to Soil testing laboratory
34. Final Practical Examination

References

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2. Garison Sposito, 2008. The Chemistry of Soils. Oxford University Press, USA
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3. www.soiltesting.okstate.edu/
4. www.texasplantandsoillab.com/
5. www.ulm.edu/spal/
6. soilhealth.cals.cornell.edu/extension/.../managing_constraints.pdf
7. cnal.cals.cornell.edu/
8. http://agritech.tnau.ac.in/agriculture/agri_reosurcemgt_soil_soilconstraints.html
9. <http://edis.ifas.ufl.edu/topicjertilization>

FSN 121 Bakery Technology (0+2)

Practicals

Introducing tools and equipments used in bakery unit. Quality evaluation of ingredients - Methods and Types of bakery products – Yeast based products (Bread and Bun), Cake, Cookies, Puffs, Rusk Doughnut and Therapeutic bakery foods. Icing and Cake Decorations - Quality evaluation of bakery products. Commercial and Cost of production of bakery products – Layout and Establishment of Bakery Unit.

Unit I: Fundamentals of Bakery Ingredients and Equipment

Identification of bakery tools and equipment. Introduction to Bakery Ingredients - flour, sugar, leavening agent, shortening agent, egg, water, colouring and flavouring agents, improvers, and preservatives. Quality Evaluation of Flour - gluten content, water absorption capacity, moisture, and ash content.

Unit II: Bread and Bun Technology

Different methods of bread preparation: straight dough, sponge-dough. Different types of bread preparation: white bread, wheat bread, milk bread, sweet bread, fruit bread, millet bread, multi-grain bread, and garlic bread. Methods of bun preparation - Straight dough and Sponge dough method, Types of bun preparation – Bun, cream bun, masala bun and stollen. Processing technology of coconut bun and butter bun.

Unit III: Cake and Icing Technology

Different methods of cake preparation - the Cream method and the all-in-one method. Types of cake preparation – Sponge cake, tea cake, angel cake, plum cake, eggless cake and cupcake. Icing- Butter Icing and Royal icing. Cake decoration- Fondants and fudge. Preparation of fresh cream cake and frosting.

Unit IV: Cookies, Pastries, and other Bakery Products

Different methods of cookies preparation- drop and sheet method. Types of cookies- Maida cookies, wheat cookies, millet cookies, composite flour cookies, coconut cookies, peanut cookies, salt cookies and chilly cookies. Method of preparation of pastry – Puff, Flaky pastry. Preparation of Naan kattai, vanilla buttons, puffs – vegetable, egg and mushroom puff; Khari – Sweet and Spices; Sweet Doughnut, Chocolate Doughnut; and Rusk – Milk and Millet Rusk.

Unit V: Quality Evaluation, Therapeutic Bakery, and Commercial Production

Quality evaluation of bakery products. Therapeutic bakery production- High fibre foods. Commercial and Cost of Production of Cakes, Bread, and Cookies. Layout and establishment of the bakery unit.

Practical Schedule

1. Identification and utilization of bakery tools and equipments
2. Quality evaluation of flour for bakery products
3. Different method of bread preparation- Straight dough and Sponge dough method
4. Types of bread preparation- White bread and Wheat bread
5. Types of bread preparation- Millet and Garlic bread
6. Different methods of bun preparation- Straight dough and Sponge dough method
7. Preparation of different types of bun- Maida bun and Cream bun
8. Preparation of different types of bun- Masala bun and Stolen
9. Different methods of cake preparation- Cream method and all in one method
10. Types of cake preparation – Sponge cake and Tea cake
11. Types of cake preparation – Angel cake and Plum cake
12. Types of cake preparation – Eggless cake and Cup cake
13. Icing- Butter Icing and Royal icing
14. Cake decoration- Fondants and fudge
15. Preparation of fresh cream cake and frosting
16. Different methods of cookies preparation- Drop method and Sheet method
17. Types of cookies- Maida and Wheat cookies
18. **Mid Semester Examination**
19. Types of cookies- Millet and Composite flour cookies
20. Types of cookies- Coconut cookies and Peanut cookies
21. Types of cookies- Salt cookies and Chilly cookies
22. Method of preparation of pastry – Puff and Flaky pastry
23. Preparation of Naan kattai and vanilla buttons
24. Preparation of puffs –Vegetable, Egg and Mushroom puff
25. Preparation of Khari – Sweet and Spices
26. Preparation of Sweet Doughnut and Chocolate Doughnut
27. Preparation of Rusk – Milk and Millet Rusk
28. Quality evaluation of bakery products
29. Therapeutic bakery production- High fiber foods

30. Commercial and Cost of production of Cakes
31. Commercial and Cost of production of Bread
32. Commercial and Cost of production of Cookies
33. Layout and establishment of bakery unit

34. Final Practical Examination

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4. www.foodindustryindis.com

FSN 122 Processing and Value-addition of Fruits and Vegetable Products (0+2)

Practicals

Unit I: Post-Harvest Processing of Horticultural Crops

Packaging technology – prepackaging by vents - waxing, vacuum packaging technology - for export by air, road and sea for fruits and vegetables. Packaging machinery and operational skill – Ripening methods – Quality assessment of packed crops for marketability.

Unit II: Processing and Value addition using Sugar and Salt

Processing using sugar - principles and processing of jam, jelly, marmalade, fruit bar, preserves and candies. Unfermented and fermented products - fruit juices, RTS, nectar, cordial, squash, syrup, carbonated beverage, wine, cider and vinegar. Processing using salt - principle – brining. Preservation of horticultural products.

Unit III: Processing of Dehydrated and Canned Products

Drying and Dehydration of fruits – mango, guava, sapota, pineapple and amla, Drying and Dehydration of vegetables – moringa, greens, bhendi, cucurbits, brinjal, gourds and other regional vegetables. Osmotic dehydration of fruits – *tutifruity*, *murraba*, glazed candy and crystalline candy. Canning of fruits – orange and pineapple, canning of vegetables – carrot and beans - Quality assessment of Products – moisture and acidity.

Unit IV: Processing of Pickles, Sauces, and Condiments

Processing and evaluation of different types of pickles – vinegar, salt & spices with and without oil. Production of tomato paste, puree, sauce, and ketchup, including red and green chilli sauce.

Unit V: Waste Utilization and Commercial Production

Utilization of fruit and vegetable waste – mango peel waste for convenience foods, apple and jack fruit for pectin production – commercial Production of jam, squash and dehydrated products.

Practical Schedule

1. Selection and sorting of raw material for Processing and preservation
2. Pre-Packing operations of fruits and vegetables – Vented Polybags and cartoons
3. Waxing, and Vacuum Packing of fruits and vegetables
4. Ripening Techniques – chemical and Natural Methods
5. Processing of Fruit Jam (Mango or Mixed Fruit)
6. Processing of Jelly- guava / apple

7. Processing of orange marmalade
8. Processing of Fruit Preserve / Candy (Amla or Pineapple)
9. Processing of Fruit Bar / Leather (Guava or Mango Pulp)
10. Processing of Unfermented Beverages –RTS and nectar
11. Processing of squash as per FSSAI specification
12. Processing of lime cordial
13. Processing of Unfermented Beverages - synthetic Syrup
14. Processing of Carbonated Fruit Beverage
15. Processing of Vinegar or Cider (Fermented Product)
16. Processing of Dehydrated Fruit Slices (Mango / Guava / Amla / Sapota / Pineapple)
17. Processing of Dehydrated Vegetables (Moringa Leaves / Bhendi / Cucurbits / Brinjal / Gourds)

18.Mid-Semester Examination

19. Osmotic Dehydration – Processing of Tutti Frutti / Glazed Candy
20. Processing of Crystalline Fruit Candy (Amla or Pineapple)
21. Canning of Fruits – Pineapple or Orange Segments
22. Canning of Vegetables – Carrot and Beans
23. Quality Assessment – Moisture, Acidity, and Sensory Evaluation
24. Processing of Brined Vegetables and Mango Pickle (with and without Oil)
25. Processing of Lime or Mixed Vegetable Pickle (Salt-Spice Pickle)
26. Processing of Tomato Puree and Paste
27. Processing of Tomato Sauce and Ketchup
28. Processing of Chilli Sauce (Red/Green)
29. Preparation of Mango Peel-Based Value-Added Product (Candy or Flour Mix)
30. Extraction of Pectin from Apple Pomace / Jackfruit Waste
31. Commercial production of Jam – single and mixed fruit
32. Commercial production of dehydrated fruits – Tuti fruiti and raisin
33. Commercial production of Squash and RTS

34.Final Practical Examination

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e- Resources

1. www.foodnetbase.com
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HOR 121 Nursery Production in Horticulture Crops (0+2)

Practicals

Unit I: Introduction to Nursery

Layout of model nursery, Tools and equipment-identification and application- Containers for propagation and their practical usage-Practices in preparation of media for propagation of fruit plants, flowering annuals and perennials, foliage and indoor ornamentals and orchids, Potting, de potting and repotting

Unit II: Sexual / Seed Propagation

Different methods of breaking seed dormancy stratification, scarification and use of plant growth regulators- Seed propagation- Extraction and storage of healthy seeds, seed bed preparation-Identification and raising of rootstocks for different fruit plants

Unit III: Understanding Nursery Components

Types of nursery beds, soil solarization- raising hybrid vegetable seedlings in Protrays, Protection of nursery plants against adverse climatic conditions- Protected structures, shade net houses and Mist chamber- Mother plant block or scion block establishment- Calculation of growth regulators concentration for propagation- Preparation and application of growth regulators for cuttings and layering,

Unit IV: Sexual / Vegetative propagation

Hands on training in preparation of stem, root, leaf and leaf bud cuttings and rooting, Different types of layering methods- Hands on training in simple layering and air layering- Rootstocks for commercial horticultural crops and raising of the rootstocks - different types of grafting methods - Hands on training in epicotyl grafting, soft wood grafting, inarching /approach grafting in fruit crops-Hands on training in shield budding and patch budding, Micro propagation- Commercial tissue culture laboratory- Visit to commercial tissue culture laboratory,

Unit V: Nursery management practices

Weed control, irrigation, nutrition, removal of sprouts etc. Diagnosis and control of important diseases and pests in the nursery- lifting and packing of nursery plants, marketing of products in commercial nursery - Nursery registration act -Obtaining accreditation for a commercial nursery - Plant quarantine, quality standards for export and import of seeds and nursery plants produced - Project preparation for the establishment of commercial nursery.

Practical Schedule

1. Layout of model nursery -Horticultural nurseries, types and economical importance and Various components of nursery
2. Tools and equipment-identification and Practices in usage and maintenance
3. Different types of media -Qualities of an ideal propagation media - Practices in preparation of media for propagation of horticultural crops
4. Containers for propagation and their practical usage
5. Practices in Potting, de potting and repotting
6. Different methods of breaking seed dormancy stratification, scarification and use of plant growth regulators and seed propagation of horticultural crops
7. Types of nursery beds – hands on training for preparation of raised beds and seed sowing
8. Hands on training raising hybrid vegetable seedlings in Protrays
9. Protected structures for propagation and
10. Hands on training for establishment of mother plant block or scion block establishment
11. Hands on training in preparation of different types of stem cuttings in horticultural crops viz., fruits and ornamental crops
12. Calculation of growth regulators concentration for propagation
13. Preparation and application of growth regulators for cuttings and layering
14. Hands on training in preparation of different types of layering in horticultural crops viz., fruits and ornamental crops
15. Hardening of rooted cuttings and layers in horticultural crops
16. Hands on training for raising the rootstocks in fruits and vegetable crops
17. Different types of grafting methods - Hands on training in epicotyl grafting, soft wood grafting, inarching /approach grafting in fruit crops
18. **Mid Semester Examination**
19. Hands on training in grafting in vegetable crops
20. Hands on training in In situ grafting and top working in fruit crops
21. Hands on training in shield budding and patch budding in fruits and ornamental crops
22. Hands on training in propagation of foliage ornamental plants
23. Hands on training in propagation of flowering ornamental plants

24. Hands on training in propagation of Orchids Anthuriums, important flowering climbers, important ornamental trees
25. Hands on training in collection and preparation of explants, surface sterilization of explants, inoculation of explants and in sub culturing
26. Hands on training in micro propagation of banana
27. Hardening of seedling in nursery
28. Nursery management practices- i.e. weed, irrigation management nutrition and removal of rootstock sprouts
29. Diagnosis and control of important diseases and pests in the nursery
30. Lifting, labelling and packing of nursery plants
31. Nursery registration act, plant quarantine, quality standards for export and import of seeds and nursery plants
32. Visit to accredited nurseries and commercial tissue culture laboratory
33. Project preparation for the establishment of commercial nursery
34. **Final practical examination**

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2. <https://iht.edu.in/basics-of-nursery-production/>
3. https://k8449r.weebly.com/uploads/3/0/7/3/30731055/horticulture_plant_nursery1-signed.pdf
4. https://msmedichennai.gov.in/GARMS_Admin/basictools/images/ProjectProfiles/FruitNursery.pdf

HOR 122 Orchard Floor Management (0+2)

Practicals

Unit I: Understanding the Components of Orchard

Types of orchards - Assessment, Designing and Layout of different systems of orchards and Fruit Nutrition Garden. Components – non-plant components of orchard.

Unit II: Soil Management and Cropping Pattern

Soil management practices: clean cultivation, sod culture, sod mulch, intercropping, cover cropping and mixed cropping.

Unit III: Irrigation System, Weed and Water Management

Use of mulch materials: organic and inorganic, moisture conservation and weed control, drainage systems. Layout of various irrigation systems, surface irrigation: Flood system, basin system, modified basin system, furrow method, sub-surface irrigation systems:

Unit IV: Micro Irrigation, Nutrient Management and Plant Protection

Layout of drip irrigation and its components, overhead irrigation: sprinkler system, Soluble fertilizers and fertigation techniques, Different methods of application of manure and fertilizers, use of organic manures, Biofertilizers, Green manuring and bio-agents. IPM, IDM practices.

Unit V: Canopy Management and Mechanization

Canopy management, crop regulation and rejuvenation techniques in fruit crops. Off season bearing techniques, Farm mechanization techniques. Calendar of operations and budget preparation. Visit to orchards of Progressive fruit growers.

Practical Schedule

1. Study of different types of orchards.
2. Assessment, data gathering and analysis of new site for the establishment of Fruit orchard.
3. Planning and designing for orchard establishment
4. Components of Orchard
5. Steps to be followed for the establishment of orchard.
6. Practices on management of physical practices of the soil.
7. Problem soil management practices
8. Study of sod culture and sod mulch
9. Study of Inter cropping in orchard with examples

10. Study of cover cropping in orchard with examples
11. Study of mixed cropping techniques with examples
12. Study of organic and inorganic mulches
13. Study of drainage systems in orchard management
14. Studies on moisture conservation practices in orchard.
15. Study of weed management practices in orchard.
16. Study of different training systems in orchard.
17. Study of wind breaks in orchards with examples

18.Mid Semester Examination

19. Study of different irrigation systems in orchards surface: flood system, basin, modified basin, furrow method etc.,
20. Designing and layout of drip irrigation systems
21. Different method of application of manures and fertilizers and calculation of fertilizer doses.
22. Study of soluble fertilizers and fertigation techniques.
23. Uses of different organic manures biofertilizers, green manuring and bioagents.
24. Integrated pest management
25. Integrated Disease management
26. Canopy management practices followed in orchard.
27. Rejuvenation of old orchards and top working techniques
28. Studies on crop regulation practices in orchard
29. Studies on non-plant components – civil works, fence line and machinery
30. Off-Season production techniques – bahar – pruning techniques – growth regulators etc.
31. Farm mechanization techniques – Power systems – tools and implements.
32. Scheduling the calendar of operations and budget preparation.
33. Visit to Fruit orchards of progressing fruit growers

34.Final Practical Examination

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2. Chattopadhyay T.K., 2012. A Textbook on Pomology (Fundamentals, Vol-I), Kalyani Publishers, New Delhi
3. Chadha, K.L. 2001. Handbook of Horticulture, ICAR, New Delhi.
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2. <https://www.slideshare.net/slideshow/orchard-floor-management/252983711>

HOR 123 Landscape Gardening (0+2)

Unit I: Basics of Gardening and Plant Propagation

Identification and use of garden tools and equipment - Identification, description and propagation of ornamental trees - Identification, description and propagation of ornamental shrubs - Identification, description and propagation of flowering annuals, climbers, creepers, and bulbous ornamentals - Identification, description and propagation of indoor plants, foliage plants and water plants - Identification and description of palms, ferns, cycads, ornamental grasses, cacti, succulents, and ground covers

Unit II: Principles and Elements of Landscape Gardening

Understanding the principles of landscape gardening -Studying the features of formal gardens- Studying the features of informal gardens- Description and designing of plant components in a garden – edges, hedges, flower beds, flower borders, arboretum, palmatum, topiary- Description and designing of non-plant garden structures – arches, bowers, pergolas, roads, walks, paths, bridges, fountains, and statues- Landscape drafting tools – dimensioning, graphic symbols, and notations

Unit III: Landscape Design and Site Planning

Site analysis, planning and designing of a home garden - Site analysis and landscape designing of gardens for public places and places of worship - Landscape planning for roads and roundabouts - Visit to public parks to study different features and styles of gardening

Unit IV: Lawns, Special Gardens, and Indoor Gardening

Identification of lawn grasses - Establishment of lawns - Maintenance of lawns - Designing and layout of special gardens – rockery, water garden - Designing and layout of special gardens – sunken garden, roof garden - Designing and layout of special gardens – vertical garden - Indoor gardening – designing and layout of terrarium/bottle garden, dish garden - Visit to community parks, ornamental plant nurseries, and golf courses

Unit V: Advanced Practices, Art Forms, and Professional Applications

Therapeutic gardens – studying the concept and types; practising activities in therapeutic gardening - Bonsai – studying the concept, principles, styles, selection of plants, containers, and tools - Practising the art of bonsai making and maintenance - Understanding the features of software used for landscape designing - CAD in landscaping – guest lecture - Practising flower arrangement and bouquet making - Value addition – practising dry flower making and

value addition with floral waste - Project preparation and working out cost estimate for designing and laying out different types of gardens

Practical Schedule

1. Identification and use of garden tools and equipment
2. Identification, description and propagation of ornamental trees
3. Identification, description and propagation of ornamental shrubs
4. Identification, description and propagation of flowering annuals, climbers and creepers, bulbous ornamentals
5. Identification, description and propagation of indoor plants, foliage plants and water plants
6. Identification and description of palms, ferns, cycads, ornamental grasses, cacti and succulents, ground covers
7. Understanding the principles of landscape gardening
8. Studying the features of formal gardens
9. Studying the features of informal gardens
10. Description and designing of plant components in a garden – edges, hedges, flower beds, flower borders, arboretum, palmatum, topiary
11. Description and designing of non-plant garden structures - arches, bowers, pergolas, roads, walks, paths, bridges, fountains and statues
12. Landscape drafting tools: dimensioning, graphic symbols and notations
13. Site analysis, planning and designing of a home garden
14. Site analysis and landscape designing of gardens for public places and places of worship
15. Landscape planning for roads and roundabouts
16. Visit to public parks to study different features and styles of gardening
17. Identification of lawn grasses
18. **Mid Semester Examination**
19. Establishment of lawns
20. Maintenance of lawns
21. Designing and layout of special gardens - rockery, water garden
22. Designing and layout of special gardens - sunken garden, roof garden

23. Designing and layout of special gardens - vertical garden
24. Indoor gardening: designing and layout of terrarium / bottle garden, dish garden
25. Visit to community parks, ornamental plant nurseries and golf courses
26. Therapeutic gardens: studying the concept and types; practising activities in therapeutic gardening
27. Bonsai - studying the concept, principles, styles, selection of plants, containers and tools
28. Practising the art of bonsai making and maintenance
29. Understanding the features of software used for landscape designing
30. CAD in landscaping - guest lecture
31. Practising flower arrangement and bouquet making
32. Value addition - Practising dry flower making and value addition with floral waste
33. Project preparation and working out cost estimate for designing and laying out of different types of gardens
34. **Final Practical Examination**

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2. <https://celkau.in/Agrienterprises/enerprise/30.Landscape%20Gardening/1.%20Landscape%20Gardening.pdf>

ARM 121 Entrepreneurial Skills for Agribusiness (0+2)

Practicals

Unit 1: Entrepreneurial and Management skills

Assessment of Entrepreneurial Skills -Exercise on Leadership skills -Time management skills - Team building Exercises -Exercise on Decision Making Skills-Exercise on Business Communication Skills-Creativity and Innovation -Innovation challenges, Ideathon, Hackathon, Boot camp, Demo Day

Unit II: Agribusiness Opportunities

Area, production, productivity, export/import of Major agricultural commodities - Mapping of major agricultural commodities – Identification of commodity/product- Identification of Agribusiness Opportunities - Hands on exercise - Market selection -Hands on exercise - Buyer identification – Segmentation and Platforms - Consumer preference for major agricultural products – Framing Interview Schedule -Quality certification - FSSAI & NPOP, AGMARK – Certification checklist

Unit III: Business Plan

Business plan – Components, types - Hands on exercise - Drafting and preparation of Business plan -Hands on exercise - Business Model canvas-Hands on exercise - Bankable project preparation- Hands on exercise - Bankable project preparation- II

Unit IV: Business Incubation and Product Promotion

Visit to incubation centre - Start-up ecosystem – Mapping key players and their interactions Hands on exercise - Registration of a sole proprietorship firm and private limited company Source of finance - Seed money, Angel investment Source of Finance - Schemes / Funds for Startups-Pitching for Funds - Guidelines and opportunities Hands on exercise - Branding, designing the Logo and Trade mark Hands on exercise - Product promotion - Hands on exercise - Digital marketing

Unit V: Networking and mentoring

Networking with EDII-TN for Innovation Voucher Programme (IVP) - Start-up India, FAME-TN and District Industries Centre (DIC) - Mentoring session with successful start-ups - Networking events with successful start-ups- Hands on business simulation games – Entrepreneurship

Practical Schedule

1. Assessment of Entrepreneurial Skills
2. Exercise on Leadership skills

3. Time management skills
4. Team building Exercises
5. Exercise on Decision Making Skills
6. Exercise on Business Communication Skills
7. Creativity and Innovation -Innovation challenges, Ideathon, Hackathon, Boot camp, Demo Day
8. Area, production, productivity, export/import of Major agricultural commodities
9. Mapping of major agricultural commodities – Identification of commodity/product
10. Identification of Agribusiness Opportunities
11. Hands on exercise - Market selection
12. Hands on exercise - Buyer identification – Segmentation and Platforms
13. Consumer preference for major agricultural products – Framing Interview Schedule
14. Quality certification - FSSAI & NPOP, AGMARK – Certification checklist
15. Business plan – Components, types
16. Hands on exercise - Drafting and preparation of Business plan, Hands on exercise - Business Model canvas
17. Hands on exercise - Bankable project preparation- I

18. Mid Semester Examination

19. Hands on exercise - Bankable project preparation- II
20. Visit to incubation centre
21. Start-up ecosystem – Mapping key players and their interactions
22. Hands on exercise - Registration of a sole proprietorship firm and private limited company
23. Source of finance - Seed money, Angel investment
24. Source of Finance - Schemes / Funds for Start ups
25. Pitching for Funds - Guidelines and opportunities
26. Hands on exercise - Branding, designing the Logo and Trade mark
27. Hands on exercise - Product promotion
28. Hands on exercise - Digital marketing
29. Networking with EDII-TN for Innovation Voucher Programme (IVP)
30. Start-up India, FAME-TN and District Industries Centre (DIC)

31. Mentoring session with successful start-ups
32. Networking events with successful start-ups
33. Hands on business simulation games - Entrepreneurship

34.Final Practical Examination

Reference

1. Innovation and Entrepreneurship: Practice and Principles by Peter F Drucker
2. Zero to one: Notes on Start-ups, or How to Build the Future by Peter Thiel
3. The Innovator's Solution: Creating and Sustaining Successful Growth by Clayton M Christensen
4. The Lean Start-up: How Today's Entrepreneurs use continuous Innovation to create radically successful business by Eric Ries

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2. https://agribsc.kkwagh.edu.in/uploads/department_course/Entrepreneurship_Development_and_Communication_Skills.pdf

**GPB 121 General Morphology and Pollination Techniques in
Agricultural Crops (0+2)**

Unit I: General morphological description of plants

Kinds of root, stem, leaf and inflorescence; Structure of flower, calyx, corolla, aestivation, androecium, gynoecium, placentation, types of fruits.

Unit II: Modes of reproduction and pollination in plants

Sexual and Asexual reproduction- Apomixis – Modes of Pollination - mechanisms enforcing self and cross pollination in crops - working out extent of natural out crossing

Unit III - Botanical description selfing, crossing techniques and hybrid seed production in cereals

Key botanical features. Observation on floral biology – anthesis and pollination – selfing – crossing techniques, varietal development, nucleus, breeder seed and hybrid seed production in rice, maize

Unit IV: Botanical description selfing, crossing techniques and hybrid seed production in millets and pulses

Key botanical features. Observation on floral biology – anthesis and pollination – selfing – crossing techniques, varietal development, nucleus, breeder seed and hybrid seed production in pearl millet and redgram

Unit V: Botanical description selfing, crossing techniques and hybrid seed production in oilseed and Cotton

Key botanical features. Observation on floral biology – anthesis and pollination – selfing – varietal development, nucleus, breeder seed and hybrid seed production sunflower and cotton

Practical Schedule

1. General morphology of root, stem and leaf and its types
2. General morphology of inflorescence, flower, fruits and its types
3. Modes of reproduction in plants - Sexual and Asexual - Apomixis
4. Mode of pollination in self- and cross-pollinated crops - Mechanisms enforcing self and cross pollination in crops - working out extent of natural out crossing Key botanical features. Observation on floral biology – anthesis and pollination – selfing – crossing techniques, varietal development, nucleus, breeder seed and hybrid seed production
5. Rice
6. Rice

7. Rice

8. Rice

9. Rice

10. Rice

11. Maize

12. Maize

13. Maize

14. Maize

15. Maize

16. Maize

17. Pearl millet

18. **Mid Semester Examination**

19. Pearl millet

20. Pearl millet

21. Pearl millet

22. Redgram

23. Redgram

24. Redgram

25. Redgram

26. Sunflower

27. Sunflower

28. Sunflower

29. Sunflower

30. Cotton

31. Cotton

32. Cotton

33. Cotton

34. **Final Practical Examination**

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5. John Joel, A., C. Vanniarajan, T.S. Raveendran, and A. Gopalan 2006. Fundamentals of Crop Botany, Directorate of ODL, Tamil Nadu Agricultural University, Coimbatore –641 003.
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SST 121 Seed Production Technology 2 (0+2)

Practicals

Unit I: Introduction and identification of seeds

Importance of quality seed and characteristics; Identification of seeds; Study of external and internal seed structures of important agricultural and horticultural crops; Study on floral biology and pollination behaviour of agricultural and horticultural crops.

Unit II: Principles and practices of seed production

Maintenance of genetic purity - procedures for seed certification - isolation distance - identification of contaminants and practicing roguing; Visit to seed production field - study on field inspection and counting procedures; Genetic tools employed in hybrid seed production of different crops; Hybrid rice - planting design - staggering, synchronization, GA application and supplementary pollination techniques, identification of pollen shedders; Hybrid maize - types of hybrids - planting design - staggering, synchronization, physiological maturity - detasseling - identification of shedding tassel, cob drying, cob sorting, shelling and identification of mechanical damage; Pearl millet - practicing jerking, identification of designated diseases; Pulses - foliar application for enhancing seed set, tendrill clipping, pre harvest sanitation spray; Groundnut - arresting flower drop, pod verification, decortication, seed recovery; Sunflower - planting design, block method, supplementary pollination, identification of pollen shedders; Castor - reproductive biology, identification of bloom types, pistillate mechanism, sex expression, foliar application; Cotton - planting design, block method, emasculation, dusting, kapas sorting, ginning and delinting; Hybrid seed production techniques in solanaceous vegetables and bhendi - seed production in cucurbits; Assessment of physiological maturity indices in different crops.

Unit III: Post harvest handling and seed storage

Practicing seed extraction and drying methods in various crops; Practicing seed cleaning, grading and upgrading techniques in various crops; specific gravity separation in paddy and bhendi; Visit to seed processing unit; Study on packaging container, seed storage godowns and maintenance. Assessment of seed dormancy and breaking methods; Seed quality enhancement techniques - seed priming, coating and pelleting.

Unit IV: Seed testing

Seed sampling: mixing, dividing and equipments; Physical purity analysis of seeds; Estimation of seed moisture content; Conducting seed germination test and seedling evaluation; Determination of seed viability; Quick viability test - seed and seedling vigour tests;

Assessment of genetic purity - Grow out test; Visit to study on farmers participatory seed production - seed hubs and community seed bank. Visit to seed industry.

Unit V: Seed production planning

Seed production planning - calculation of area requirement for different classes of seeds and working out economics of seed production.

Practical Schedule

- 1 Importance of quality seed and characteristics - Identification of seeds
- 2 Study of external and internal seed structures of important agricultural and horticultural crops
- 3 Study on floral biology and pollination behaviour of agricultural and horticultural crops
- 4 Maintenance of genetic purity - procedures for seed certification - isolation distance - identification of contaminants and practicing roguing
- 5 Visit to seed production field - study on field inspection and counting procedures
- 6 Genetic tools employed in hybrid seed production of different crops
- 7 Hybrid rice - planting design - staggering, synchronization, GA application and supplementary pollination techniques, identification of pollen shedders
- 8 Hybrid maize - types of hybrids - planting design - staggering, synchronization, physiological maturity - detasseling - identification of shedding tassel, cob drying, cob sorting, shelling and identification of mechanical damage
- 9 Pearl millet - practicing jerking, identification of designated diseases
- 10 Pulses - foliar application for enhancing seed set, tendrils clipping, pre harvest sanitation spray
- 11 Groundnut - arresting flower drop, pod verification, decortication, seed recovery
- 12 Sunflower - planting design, block method, supplementary pollination, identification of pollen shedders
- 13 Castor - reproductive biology, identification of bloom types, pistillate mechanism, sex expression, foliar application
- 14 Cotton - planting design, block method, emasculation, dusting, kapas sorting, ginning and delinting

- 15 Hybrid seed production techniques in solanaceous vegetables and bhendi
- 16 Seed production techniques in cucurbits
- 17 Assessment of physiological maturity indices in different crops - practicing seed extraction and drying methods in various crops
- 18 **Mid Semester Examination**
- 19 Practicing seed cleaning, grading and upgrading techniques - specific gravity separation in paddy and bhendi
- 20 Visit to seed processing unit
- 21 Study on packaging container and seed storage godown and maintenance
- 22 Assessment of seed dormancy and breaking methods
- 23 Seed quality enhancement treatments - seed priming, coating and pelleting
- 24 Seed sampling: mixing, dividing and equipments
- 25 Physical purity analysis of seeds
- 26 Estimation of seed moisture content
- 27 Conducting seed germination test and seedling evaluation
- 28 Determination of seed viability - Quick viability test
- 29 Seed and seedling vigour tests
- 30 Assessment of genetic purity - Grow out test
- 31 Visit to seed production fields - study on farmers participatory seed production, seed hubs and community seed bank
- 32 Visit to seed industry
- 33 Seed production planning - calculation of area requirement for different classes of seeds and working out economics of seed production

34 Final Practical Examination

References

1. Agarwal, P.K. and M. Dadlani, 1987. Seed Science and Technology, South Asia Publishers, New Delhi.
2. Agarwal, R.L., 1996. Seed Technology, Oxford and IPH Publicity Company, New Delhi.

3. Bale, M.S., 2013. Seed Certification Handbook, Vardhman Books and Periodicals.
4. Joshi, A.K. and Singh, P.D., 2003. Seed Science and Technology, Kalyani Publishers, Ludhiana.
5. Kulkarni, G.N., 2002. Principles of Seed Technology, Kalyani Publishers, Ludhiana.
6. Nema, N.P., 1986. Principles of Seed Certification and Seed Testing, Allied Publishers Private Limited. New Delhi.
7. Singh. P.D. 2018. Principles and Methods of Plant Breeding. Kalyani Publishers, Ludhiana.

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2. <https://pmc.ncbi.nlm.nih.gov/articles/PMC4233836/>
3. <https://sites.google.com/a/uasd.in/ecourse/seed-science-technology>

AMP 121 Broiler Production (0+2)

Practical

Unit I: Introduction

Poultry Industry in India - Current status of broiler industry – Scope of broiler production in India - Introduction to Broilers – Commercial strains of broilers

Unit II: Housing Management

Housing management – Location and layout of commercial broiler farm – Preparation of poultry house - Equipments used in broiler farm - Different system of Management - Deep litter system, Cage system of management, Raised housing - Litter management - Preparation of brooder house – Brooder Management – Broiler breeder - Chick management – Grower management – Layer management - Summer management of broiler -Winter management of broiler – Lighting management – Common procedures followed in broiler farm.

Unit III: Feeding Management

Water – Requirement, quality analysis and its maintenance - Feeding Management of broilers – Types of feed – Feed ingredients – Quality assessment of feed ingredients and feed – Additive and supplementation of additives - Storage of feed-Feeding Methods – Nutrient requirement of different stages of broilers – Various standards for broilers -Feed formulation – Least cost formulation –Preparation of compound feed –Components in feed mill –Feed mill operations

Unit IV: Flock Health Management

Common disease of broilers – Control and Prevention - Medication and Vaccination in broilers – Different vaccination methods – Cold chain for vaccine – Vaccination schedule for broilers - Postmortem inspection –Waste management - Disposal of dead birds and Manure management - Biosecurity measures.

Unit V: Hatchery Management, Processing and Marketing

Hatchery Management – Hatchery layout and design – Hatcher and Setter – Collection and handling of egg – Setting and hatching of eggs and chicks quality assessment - Processing of broilers - Slaughtering of broilers and cut up parts of broilers - Record maintenance- Marketing Channels in broilers - Export of poultry meat - Integration method of broilers marketing– Economics of broiler farming and Project preparation for broiler farm unit for bank loan–Role of NECC and BCC in marketing of poultry and its products - Visit to commercial broiler farm feed plant, hatchery unit and processing plant.

Practical Schedule

1. Current status of broiler industry – Scope of broiler production in India - Commercial strains of broilers
2. Location and layout of commercial broiler farm
3. Preparation of poultry house
4. Equipments used in broiler farm
5. Different system of Management - Deep litter system, Cage system of management, Raised housing – Video on broiler management.
6. Litter management
7. Preparation of brooder house – Brooder Management,
8. Broiler breeder - Grower management and layer management
9. Summer and winter management of broiler
10. Lighting management – Common procedures followed in broiler
11. Water – Requirement, quality analysis and its maintenance
12. Types of feed – Feed ingredients
13. Feeding Management of broilers
14. Quality assessment of feed ingredients and feed - Additive and supplementation of additives
15. Storage of feed-Feeding Methods
16. Nutrient requirement of different stages of broilers – Various standards for broilers
17. Feed formulation – Least cost formulation
18. **Mid Semester Examination**
19. Preparation of compound feed
20. Components in feed mill – Feed mill operations
21. Common bacterial disease of broilers – Control and Prevention
22. Common viral disease of broilers – Control and Prevention
23. Common metabolic and fungal disease of broilers – Control and Prevention
24. Common deficiency disease of broilers – Control and Prevention
25. Medication and Vaccination in broilers and layers – Different vaccination methods – Cold chain for vaccine – Vaccination schedule for broilers
26. Postmortem inspection - Disposal of waste -dead birds and manure

27. Biosecurity measures.
28. Hatchery Management – Hatchery layout and design – Hatcher and Setter –
29. Collection and handling of egg – Setting and hatching of eggs and chicks quality assessment
30. Processing of broilers - Slaughtering of broilers and cut up parts of broilers – Video on modern slaughtering of broilers.
31. Record maintenance- Marketing Channels in broilers - Export of poultry meat
32. Integration method of broilers marketing
33. Economics of broiler farming and Project preparation for broiler and layer farm unit for bank loan Role of NECC and BCC in marketing of poultry and its products & Visit to commercial broiler farm, feed plant, hatchery unit and processing plant
34. **Final Practical Examination**

References

1. *Poultry Science* by Colin G. Scanes and Karen D. Christensen. This comprehensive text is widely used at the university level and covers global and commercial production of meat chickens (broilers), genetics, diseases, housing, and processing.
2. *Textbook on Commercial Poultry Production and Hatchery Management* by M. Murugan. This is a detailed publication, particularly useful for commercial operations and academic settings, covering all aspects of production and hatchery management.
3. *Commercial Broiler Chicken Production*. This title is offered by multiple publishers (e.g., IBDC, The Bookish Owl) and focuses on the specific practices of large-scale broiler meat production.
4. *Poultry Science and Practice: A Textbook* by N. Ghosh. A general textbook on poultry science that includes essential information on broiler production.
5. *Modern Livestock and Poultry Production* by James R. Gillespie and Frank B. Flanders. A broad textbook that covers current practices in livestock and poultry production, including dedicated sections on broilers

e- Resources

1. http://www.agritech.tnau.ac.in/expert_system/poultry/Broiler%20Management.html
<https://pmc.ncbi.nlm.nih.gov/articles/PMC4233836/>
2. <https://sites.google.com/a/uasd.in/ecourse/seed-science-technology>

PAT 121 Commercial Mushroom Production (0+2)

Practicals

Unit I: Mushroom current status, media and tissue culture

Mushroom - Introduction, importance-present production and trade, scope for export, markets for mushroom and mushroom products Different types of media - Tissue isolation- Equipment's required for culture media and spawn preparation

Unit II: Oyster mushroom – spawn production, cultivation, biotic and abiotic disorders

Oyster mushroom - mother spawn and bed preparation - Oyster mushroom cultivation - cropping house requirements - pest and diseases and abiotic disorder - Visit to oyster mushroom farm

Unit III: Milky mushroom – spawn production, cultivation, biotic and abiotic disorders

Milky mushroom - mother and bed spawn preparation - milky mushroom cultivation - cropping house requirements-bed preparation and casing - pest, disease and abiotic disorders - Visit to Milky mushroom farm

Unit IV: Button mushroom – spawn production, cultivation, biotic and abiotic disorders

Button mushroom cultivation - visit to button mushroom unit-compost preparation- cropping house requirements- pest, disease and abiotic disorders

Unit V: Value addition and Project preparation

Recipe and value added products from oyster, milky and button mushroom - Mushroom as a component in Integrated Farming System - Project preparation on oyster, milky and button – mushroom production and economics

Practical Schedule

1. Importance of mushroom - Status of edible and medicinal mushroom industry in India
2. General characters of mushrooms - edible and poisonous mushrooms
3. Morphological characters of edible mushroom - *Pleurotus* , *Agaricus*, *Volvariella* and *Calocybe*
4. Varieties of edible mushrooms and their special features
5. Equipments for spawn and mushroom production and their handling
6. Preparation of culture media for isolation of mushroom fungi
7. Isolation and pure culture techniques of mushroom fungi

8. Preservation and maintenance of pure culture of mushroom fungi
9. Oyster mushroom - mother spawn preparation
10. Oyster mushroom- bed spawn preparation
11. Spawn contaminants and their management
12. Cropping house for oyster mushroom cultivation
13. Substrate preparation for oyster mushroom cultivation
14. Bed preparation, maintenance, harvest and storage of oyster mushroom
15. Pest and disease of oyster mushroom- Identification and their management
16. Visit to commercial oyster mushroom unit
17. Mother spawn and bed spawn preparation for milky mushroom

18. Mid Semester Examination

19. Cropping house for milky mushroom cultivation
20. Substrate preparation for milky mushroom cultivation
21. Bed preparation for milky mushroom cultivation
22. Preparation of casing soil and its application for milky mushroom cultivation
23. Bed maintenance, harvest, grading, packing and storage of milky mushroom
24. Pest and diseases of milky mushroom - Identification and their management
25. Abiotic disorders of oyster and milky mushrooms
26. Compost preparation and its quality for button mushroom cultivation
27. Cropping room, bed preparation and cropping technology for button mushroom
28. Harvesting, grading, packing and storage of button mushroom
29. Identification of pest, diseases and abiotic disorders of button mushroom and their management
30. Recipes and value-added products from edible mushrooms
31. Visit to commercial milky and button mushroom units
32. Visit to uzhar sandai and retail shops and studying the marketing strategy for edible mushroom
33. Project preparation on oyster, milky and button mushroom production and cost economics

34. Final Practical Examination

References

1. A.S.Krishanmoorthy.,T.Marimuthu. and S.Nakkeeran.2005. Mushroom Biotechnology, Vijay Books, Sivakasi,India,Publ: ODL, TNAU, Coimbatore
2. G. Thirubhuvanamala, K. Angappan, M. Karthikeyan, P. Mahalakshmi, P. Geetha and M. Shanthi. 2024. Handbook of Oyster, Milky, Paddy straw mushroom Cultivation. Department of Plant Pathology, Coimbatore.
3. Gogoi, Robin, Y. Rathaiah, and Tasvina Rahman Borah.2019.Mushroom cultivation technology. Scientific Publishers, Jodhpur.
4. Manjith Singh, Bhuvneshvijay,Shwetkamal and G..Wakchaure. 2011.Mushrooms- Cultivation, Marketing, and Consumption, Directorate of Mushroom Research, Solan.

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1. <https://nhb.gov.in/pdf/Cultivation.pdf>
2. https://tnagriculture.in/dashboard/CPG/12_Mushroom_Cultivation.pdf
3. <https://smallfarms.cornell.edu/projects/mushrooms/methods-of-commercial-mushroom-cultivation-in-the-northeastern-united-states/4-four-methods-of-commercial-cultivation/>

COM 124 Database Management and its Application in Agriculture (0+2)

Practicals

Unit I: Introduction to Database Concepts

Fundamentals of Databases – Types of Database Models (Hierarchical, Network, Relational, ER Model) – Database Design Principles – Identification of Database Applications in Agriculture – Installation and Exploration of MS Access/MySQL – Creating a Simple Crop Database – Table Design and Primary Keys – Establishing Relationships – Data Entry and Validation Techniques.

Unit II: Database Design and Modelling

Entity-Relationship (ER) Modelling – Designing Relational Schemas – Drawing ER Diagrams for Crop, Soil and Pest Data – Introduction to Normalization – Implementation of Normalized Tables using MS Access/MySQL.

Unit III: Data Management and Querying

Overview of DDL Queries – DML Queries – Aggregate Functions – Grouping Data using GROUP BY – Joining Tables to Retrieve Related Data – Exporting Query Results to Excel.

Unit IV: User Interface Development

Introduction to Forms and reports generation - Importing/exporting data - Creating user interfaces - Designing forms for data entry - Generating reports - Setting up database backup procedures - Restoring data from backups - Securing the database.

Unit V: Case Studies and Real-world Applications

Designing a Database for Pest Management Records – Developing Soil Fertility Management Database – Integrating Weather Data with Crop Growth Database – Database for Irrigation Scheduling.

Practical Schedule

1. Course Introduction and Importance of Database Management in Agriculture
2. Introduction to Database Concepts
3. Types of Database Models and Agricultural Applications
4. Installation and Overview of MS Access/MySQL
5. Creating a Simple Crop Database
6. Designing Tables and Primary Keys
7. Establishing Relationships between Tables
8. Data Entry and Validation Techniques
9. Introduction to ER Modelling
10. Drawing ER Diagram for Agricultural Data
11. Designing Relational Schema

12. Introduction to Normalization
13. Implementing Normalized Tables
14. Overview of DDL Queries
15. DML Queries (INSERT, UPDATE, DELETE)
16. Aggregate Functions in SQL
17. Grouping Data using GROUP BY
18. **Mid Semester Examination**
19. Joining Tables for Agricultural Data
20. Exporting Query Results to Excel
21. Introduction to Forms design
22. Introduction to Reports generation
23. Importing data for forms and report
24. Exporting data from forms and report
25. Creating user interfaces
26. Designing forms for data entry
27. Generating reports
28. Setting up database backup procedures
29. 29 Restoring data from backups
30. Securing the database
31. Designing a database for pest management records
32. Developing a soil fertility management database
33. Designing weather data and irrigation scheduling database
34. **Final Practical Examination**

Reference

1. Elmasri, R., and S. B. Navathe. *Fundamentals of Database Systems*. Pearson.
2. Coronel, Carlos, and Steven Morris. *Database Systems: Design, Implementation, and Management*. Cengage Learning.
3. Oppel, Andy. *Databases: A Beginner's Guide*. McGraw Hill.
4. Silberschatz, Abraham, Henry Korth, and S. Sudarshan. *Database System Concepts*. McGraw Hill.
5. Ramakrishnan, Raghu, and Johannes Gehrke. *Database Management Systems*. McGraw Hill.
6. Date, C. J. *An Introduction to Database Systems*. Pearson.

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1. <https://www.w3schools.com/sql>
2. <https://dev.mysql.com/doc>
3. <https://learn.microsoft.com>
4. <https://www.coursera.org>
5. <https://www.skillindiadigital.gov.in>